

Tinsae Melkamu

CSE student | Back-end developer (Node.js, JavaScript, MongoDB)

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PROFILE

Computer science and engineering student specializing in backend development, with hands-on experience designing and implementing RESTful APIs using Node.js, Express, and TypeScript. Strong focus on authentication workflows, database modeling, and writing clean, maintainable, and production-style backend code through real-world projects.

EXPERIENCE & PROJECTS

DevArcAi (MVP) - AI System Design Assistant

Node.js, Express, ChromaDB, Google Gemini API, SQLite 📄

- Selected for CSEC-ASTU Development Division based on this 5-day MVP; demonstrated rapid backend architecture and AI integration skills.
- Orchestrated the AI pipeline using the Google Gemini API and ChromaDB, enabling the system to retrieve and summarize insights from technical documentation for junior developers.
- Developed a robust RESTful API using Node.js and Express to manage document ingestion, vector storage, and SQLite-based user management.
- Implemented persistent chat logic and secure admin endpoints to facilitate document uploads and real-time architecture guidance.
- Leveraged AI-assisted prototyping (Vo) to generate a responsive Next.js frontend, allowing for a primary focus on core backend logic, API integration, and vector database optimization.

Book Store Management System

TypeScript, Node.js, MongoDB 📄

- Built a type-safe RESTful API using Node.js, Express, and TypeScript, supporting CRUD operations for users and books
- Designed MongoDB schemas with Mongoose, enforcing validation and structured data modeling
- Implemented JWT-based authentication and middleware for protected routes
- Centralized error handling and request validation to improve API reliability and maintainability
- Managed version control using Git with branching for feature development

Secure Image upload API

Node.js, Express, MongoDB, Cloudinary 📄

- Built a secure RESTful API supporting user auth (signup, signin, changepassword) and image management (upload, list, delete).
- Implemented JWT authentication, bcrypt password hashing, and role-based access control via middleware.
- Designed Mongoose schemas with validation to ensure consistent data modeling.
- Integrated Multer for multipart handling and Cloudinary for scalable image storage.
- Centralized error handling and request validation to improve reliability and maintainability.
- Used Git with feature-branch workflow for version control and incremental development.

EDUCATION

BSc in Computer Science and Engineering

2023 – 2028

Adama Science and Technology University

Relevant Coursework:

- Data Structures & Algorithms
- Database Management Systems (SQL, relational modeling)
- Software Engineering (SDLC, documentation practices)

TECHNICAL SKILLS

Languages

- JavaScript(ES6+)
- TypeScript
- Java

Back-end

- Node.js
- Express
- REST architecture
- CRUD operations
- JWT authentication

Databases

- MongoDB|mongoose
- MySQL
- SQLite
- ChromaDB

Tools

- Version control(Git)
- postman